



LAND OF THE CURIOUS



EVENT AND DATE

LECTURE 8 – PRODUCT REFINEMENT
LECTURE 9 – SERVICE DEVELOPMENT
CS39A0040 Product & Service Development

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AGENDA FOR LECTURE

- Lecture 8 – Product Refinement
 - » Product Refinement (System Refinement)
 - » Production Refinement
 - » Post-Release Refinement
- » Lecture 9 – Service Development

”

” Stay hungry. Stay foolish.”

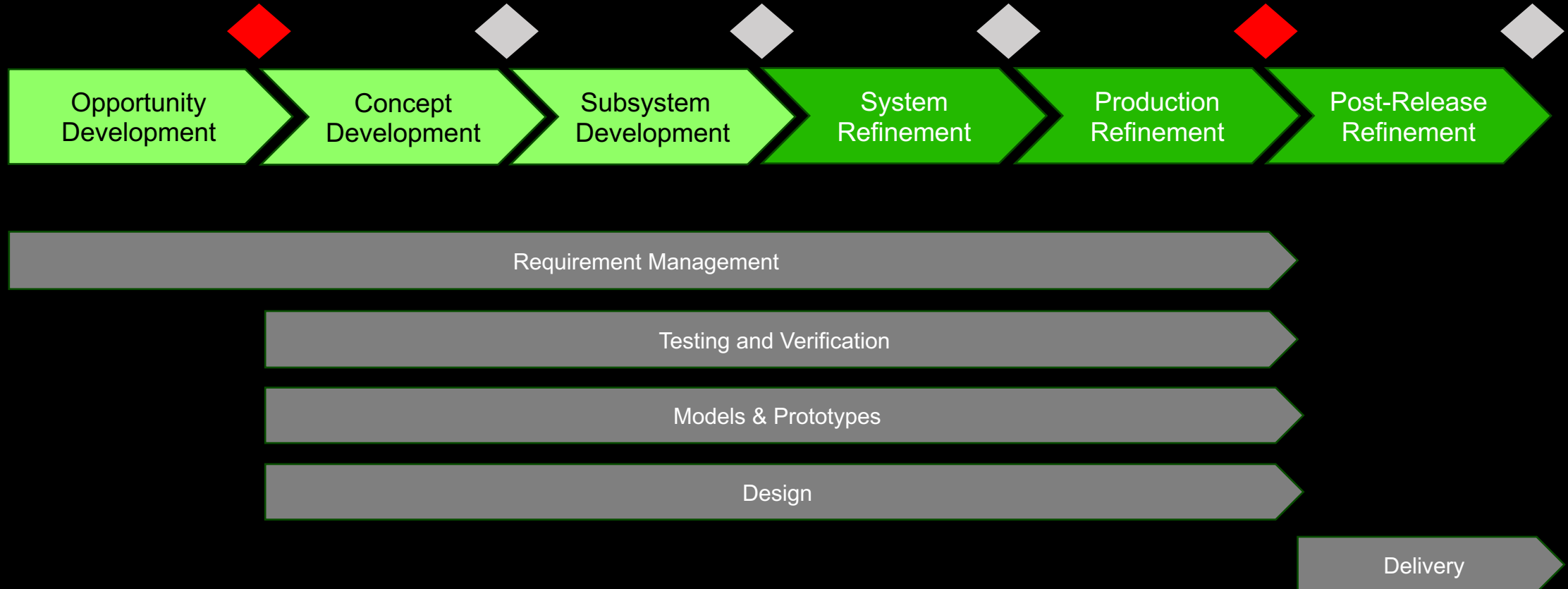
Steve Jobs





PRODUCT REFINEMENT- THE BIG PICTURE

PRODUCT DEVELOPMENT







PRODUCT AND PRODUCTION REFINEMENT

Subsystem Modules



*Major releases and
changes to product*



Changes to production





PRODUCT & PRODUCTION REFINEMENT PHASES

Release products, assemblies & production is continuously improved

READINESS LEVELS OF PRODUCTS & SERVICES

Readiness & Decisions per Market, separate dates



Artefacts and data
 Organization (PMs)
 Skills and training
 Business process rules
 KPI and bonus plans
 Information systems
 Physical Assets
 Agreements

Follow up, criteria

- » Vast amount of documentation and plans: outputs from development process
- » Some is market specific due regulation/standards and language requirements



CONTINUOUS DEVELOPMENT & IMPROVEMENT

INTRODUCTION

System refinement:

- improving weaknesses identified when the entire system is tested as a system.

Producibility refinement:

- improving product (MBOM) weaknesses observed during the start of production.

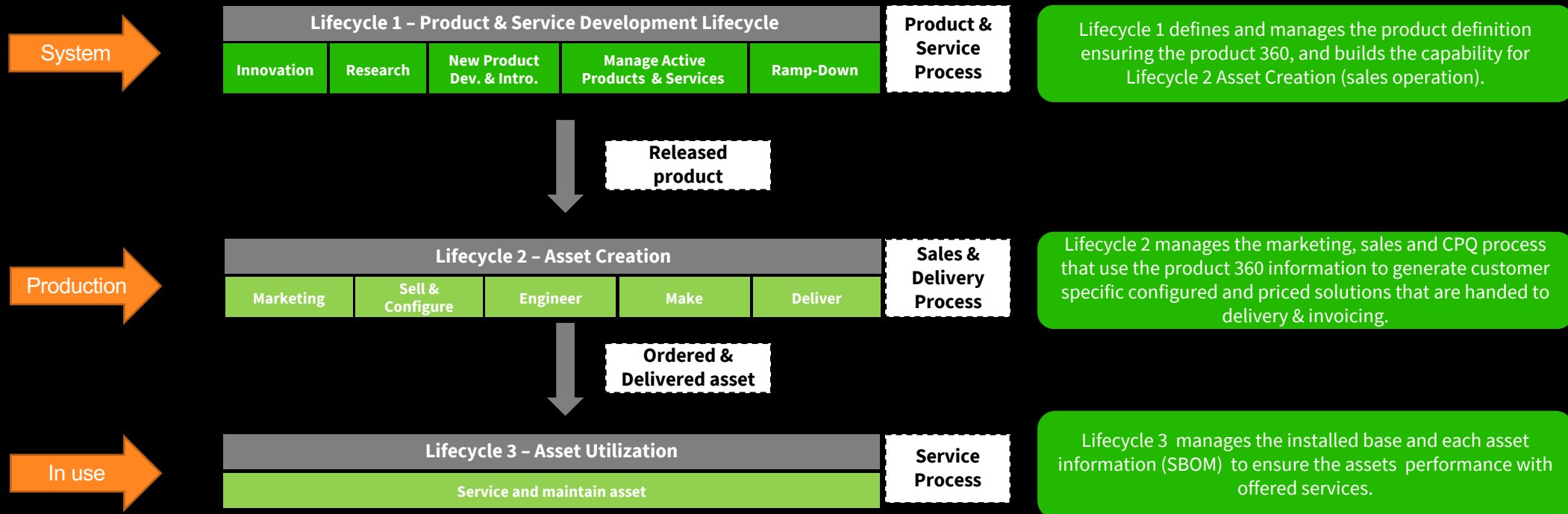
Post-release refinement:

- improving product weaknesses found by the market place, after the product was released.

OBJECTIVE OF REFINEMENT

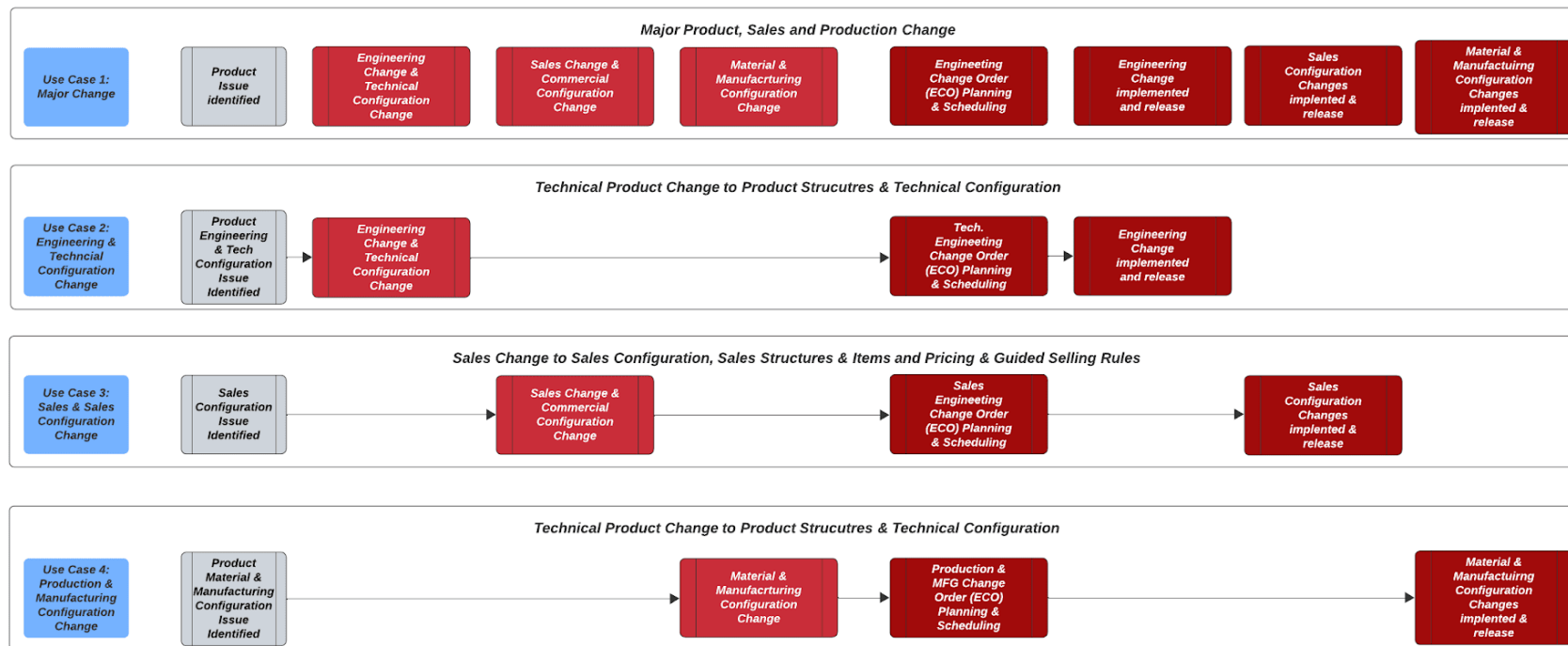
- At the end of subsystem engineering, the design is complete and released.
- Approved designs are released to other parts of the organization that will use the design data.
- Approved revisions are then conveyed to all those who need it (Rev. 1.0, 1.1., 2.1).

PRODUCT AND SERVICE COMPETITIVENESS OVER THE LIFECYCLE



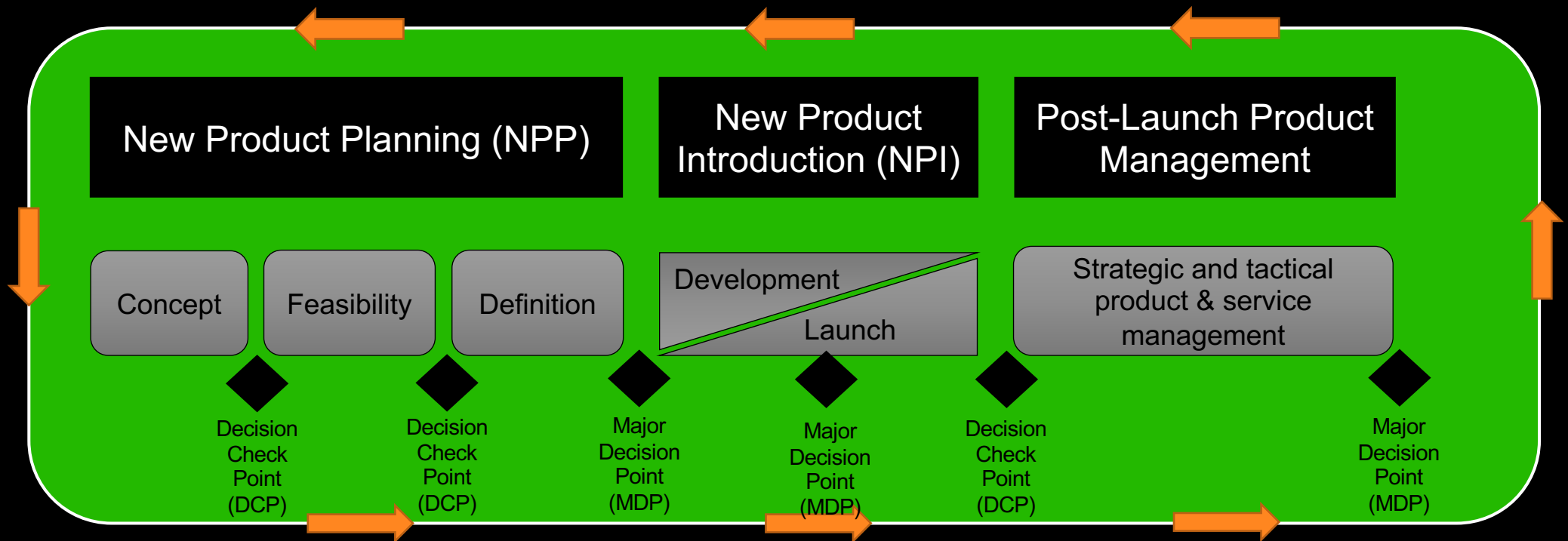
CHANGE MANAGEMENT IS CRUCIAL FOR THE SUCCESS OF MAINTAINING PRODUCT AND SERVICE CONFIGURATIONS

RELEASE AND CHANGE MANAGEMENT PROCESSES

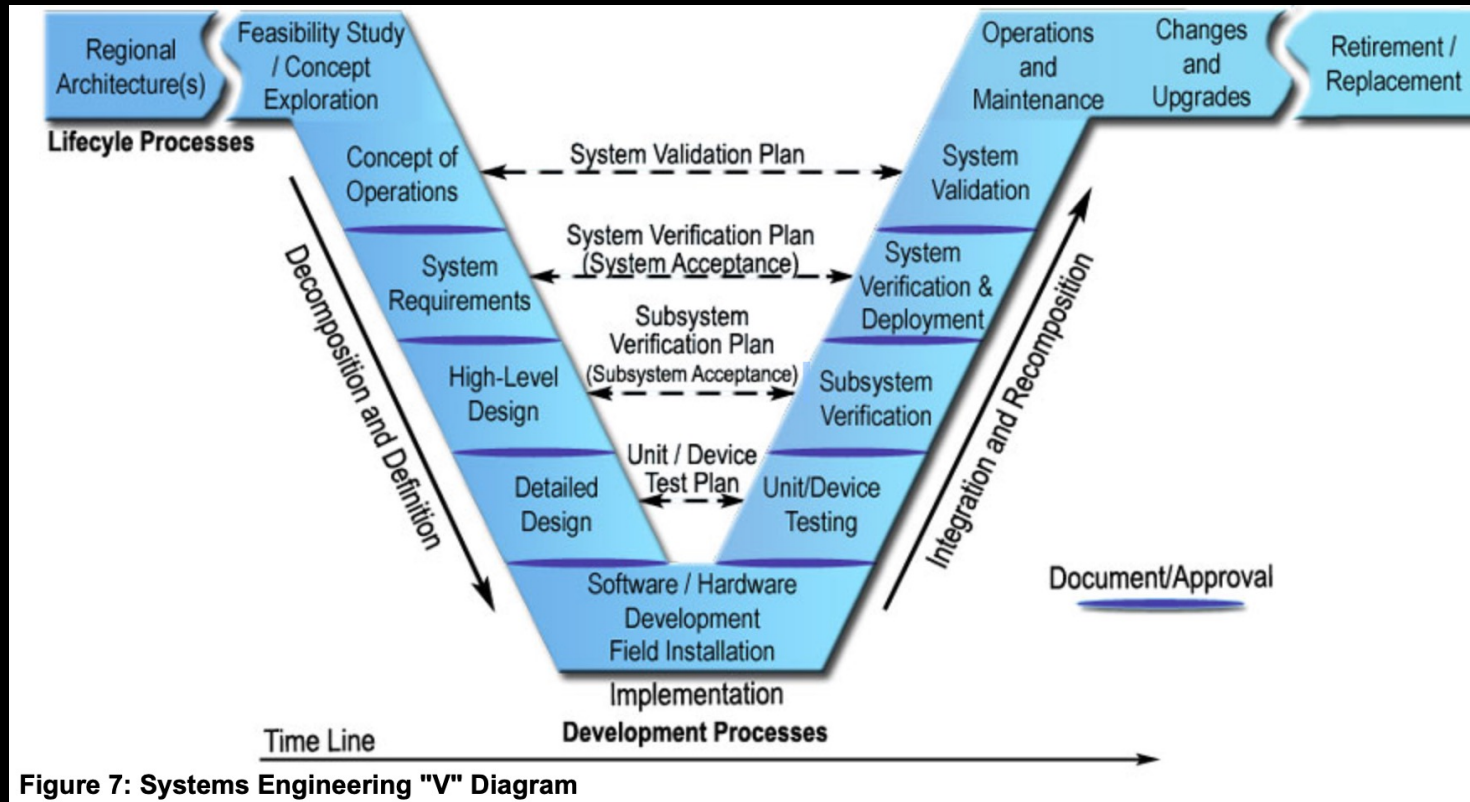


Revision control starts latest at milestones for release information

GENERIC PLANNING & DEVELOPMENT PROCESS



DEVELOPMENT PROCESS EXAMPLES



Changes are under revision control:

- Engineering Change Request (ECR)
- Engineering Change Order (ECO)
- Non-Conformance Request (NCR) (Waiver) for unplanned production changes
- Case, Problem, Warranty

SYSTEM REFINEMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Requirements	Measured system performance values	Section E of the system requirements matrix	- Are all predicted values present, even if the value is N/A? - Are all measured values present?	- System meets or exceeds the target values of the system performance measures. - If a few of the targets are not met, performance is at least in the acceptable range.
	Market response to the product	- Section F of the requirements matrix. - Reports on customer response to the product, as measured by surveys, focus groups, interviews, or other direct interaction.	Has the market response been adequately assessed?	The market finds the product desirable.
Tests	Updated methods used to predict and measure the performance of the entire system (product/service).	- Reports on methods and results demonstrating the desirability of the product. - Plots showing variation of system performance with changes in design parameters.	- Are the test methods complete and correct? - Is there enough detail for a third party to repeat the tests?	The tests have been carried out with sufficient quality and transferability to provide strong evidence for the measured and predicted values.
Models	Engineering models used to choose design parameters and predict values for system performance measures.	- Model source code with run results. - Input files for commercial software with run results. - Excel spreadsheets. MathCAD worksheets. Hand solutions.	Are the models reported in enough detail to allow a third party to use them?	Not approved directly; used with tests.

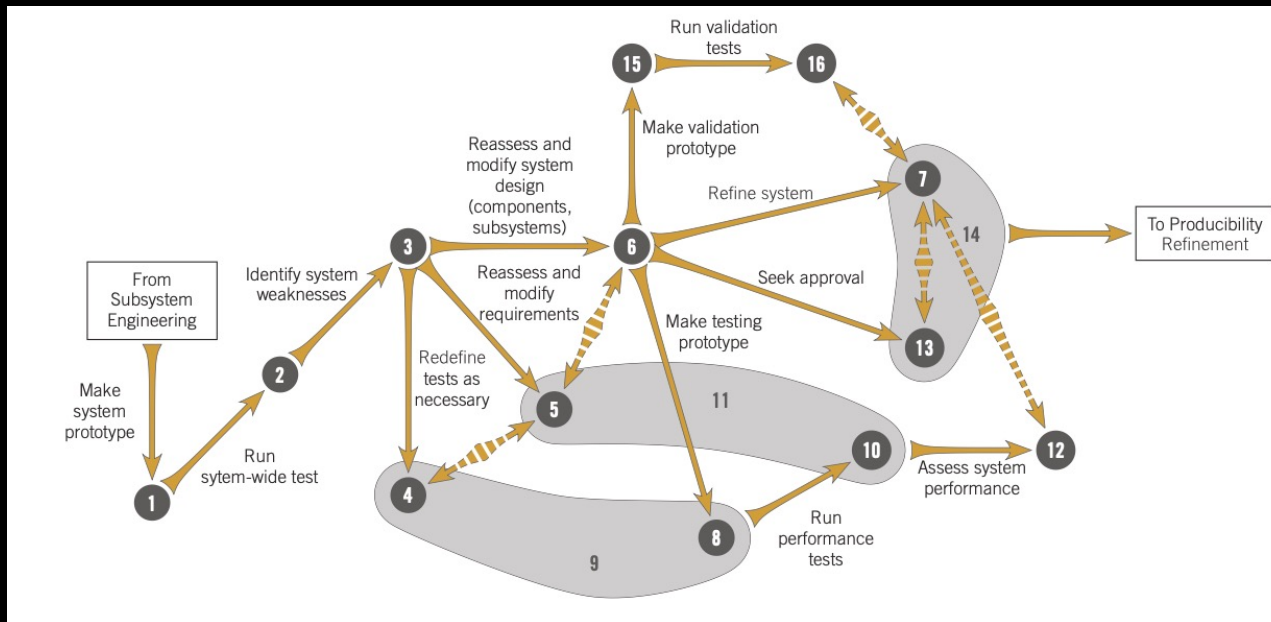
SYSTEM REFINEMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Prototype	Prototypes used to choose values of design parameters. Prototypes to determine measured values of system performance measures.	The design is sufficiently transferable to support the creation of the entire system by a third party.	- Are the prototypes appropriate for the intended use? - Is the fidelity and workmanship appropriate?	Not approved directly; used with tests.
Design	Refined definition for the entire system.	- Engineering drawings of all custom-designed parts. - Specifications (and possibly ordering information) for all purchased parts. - Subassembly & assembly drawings and instructions for all subsystems & system. - Schematic diagrams. - Piping and/or wiring diagrams. - Block diagrams. - PC board layout files. - Flowcharts & source code for any software	- Does the design package meet the design intent of the team? - Are all relevant standards met? - Are all the necessary components included? - Is the design package sufficient to allow a third party to correctly make the product and test its compliance with specifications? - Are all elements under revision control?	The design is sufficiently transferable to support the creation of the entire system by a third party.
	Complete bill of materials	Spreadsheet or database table that lists all known components	- Is it complete? - Does it have all necessary information? - Is it clear and unambiguous?	The bill of materials is appropriate

PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
	Tools:	• Bill of materials, CAD modelling, checking drawings • Design for assembly, design for manufacturing, design of experiments, design structure matrix, • Dimensional analysis, engineering drawings, experimentation, failure modes and effects analysis, fault tree analysis • Finite element modelling, prototyping, sensitivity analysis, uncertainty analysis.		
	Common Pitfalls:	• Following poor experimental procedure • Creating new and distracting performance measures • making poor trade-off; • Creating new problems • Substituting team judgment for the market.		

ACTIVITY MAP FOR SYSTEM REFINEMENT



Based on sub-system engineering

Outcomes from system refinement phase:

- Run system prototype tests
- Identify system weaknesses
- Change systems to improve performance
- Run validation tests with validation prototype
- Assess system performance

MOVE TO PRODUCTION REFINEMENT

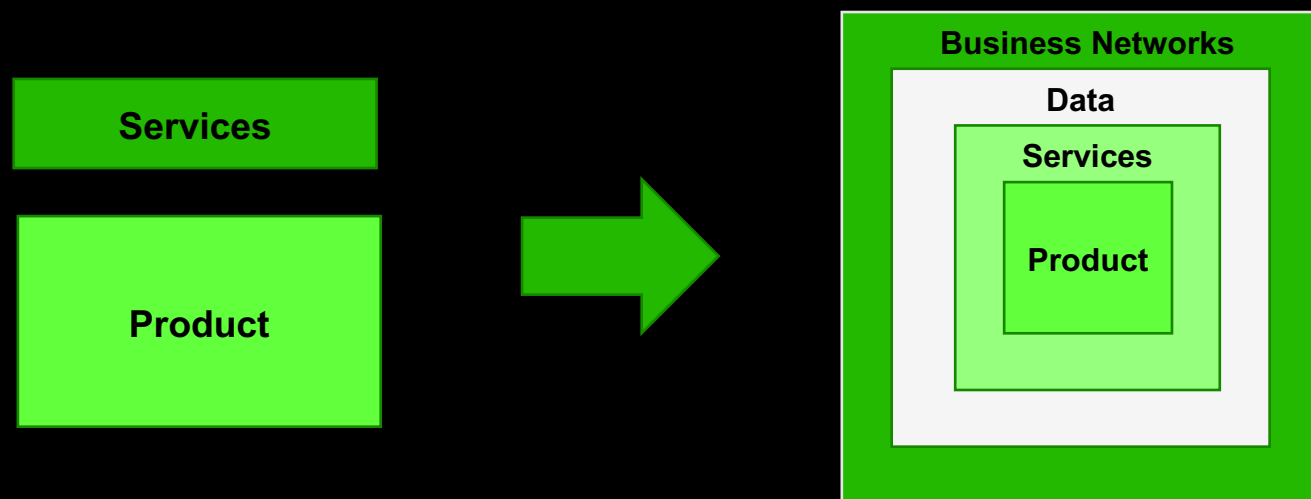
Production and pre-release follow the same procedures



SERVICE DEVELOPMENT

Release products, assemblies & production is continuously improved

PRODUCT OFFERING IS SHRINKING



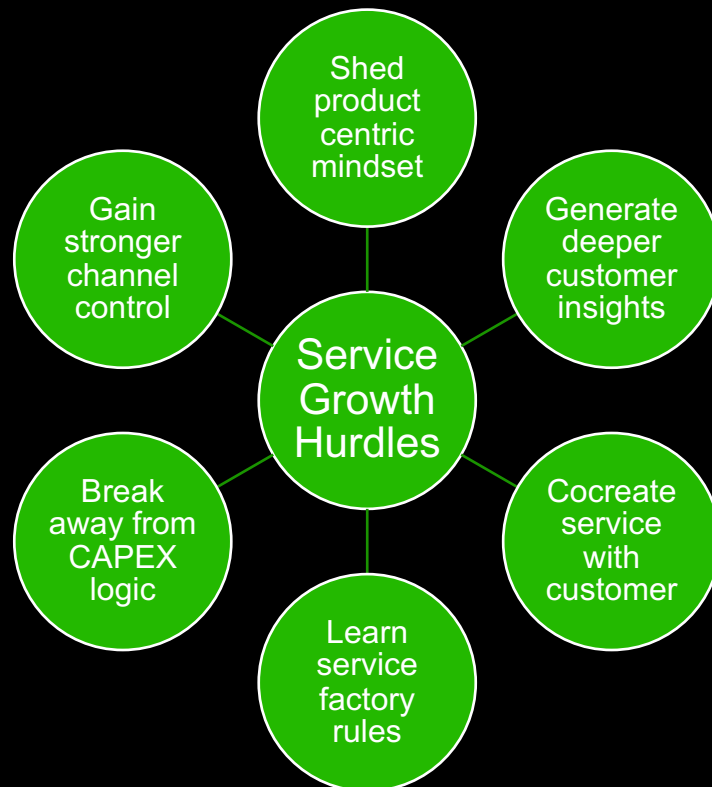
What is a company offering and has it changed?

Product, Services, PSS, iPSS?

Servitization (transition) as a **transition in business model from products to PSS**, where product and services are bundled to generate higher use-value, pricing is based on value, and capabilities support customer-dominant orientation.

(Kohtamäki et al., 2018)

MAIN HURDLES FOR TRANSITION



Smart Connected Product-Service Systems offer integrated services that change the customer relationship.

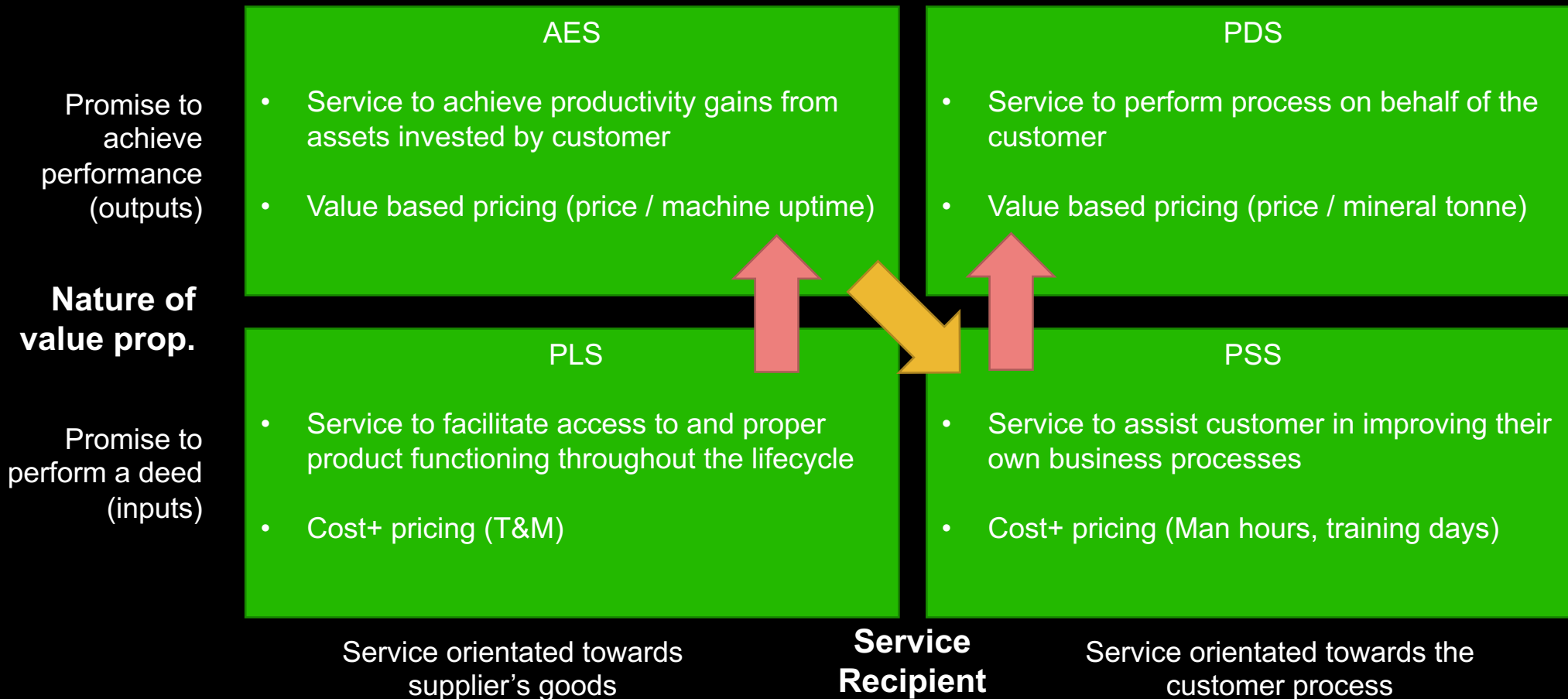
Need to understand the customer business

Partnership vs. customer

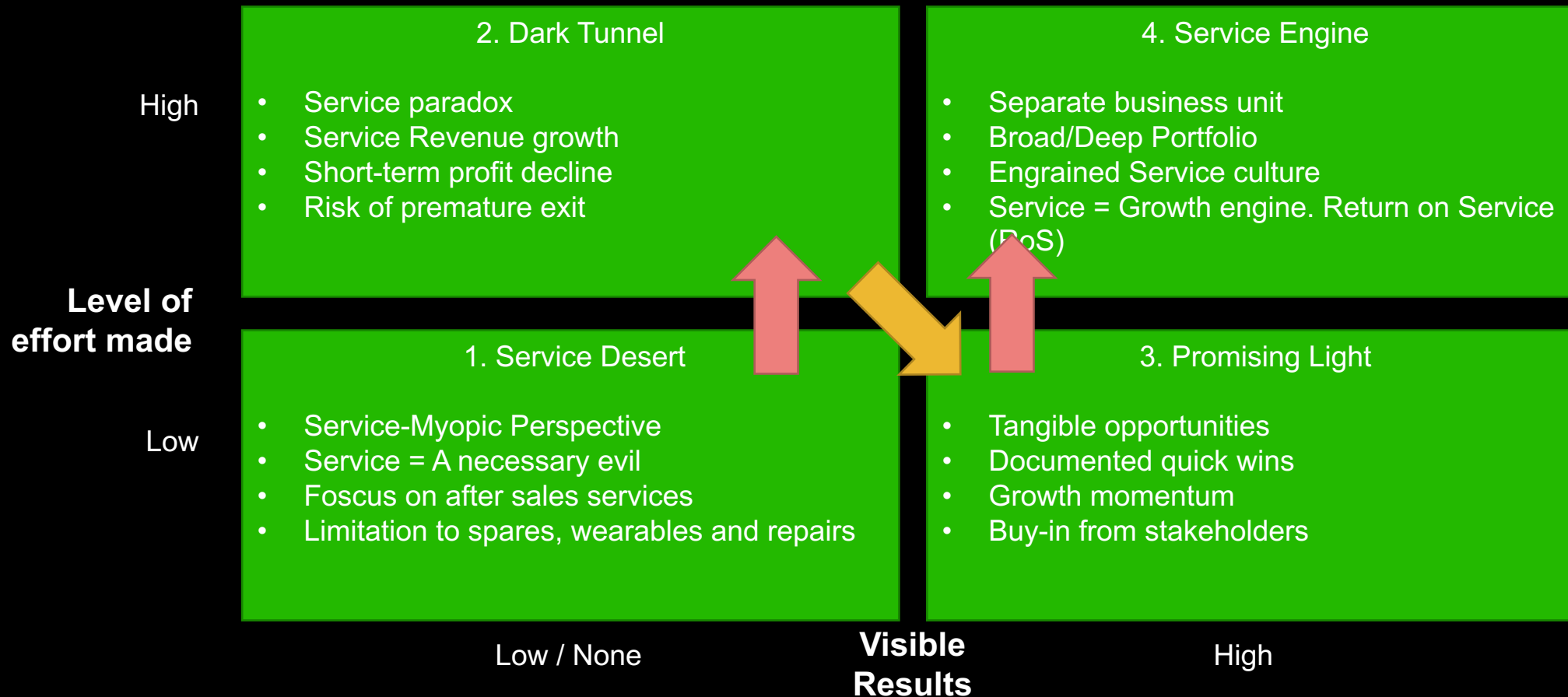
INDUSTRIAL GOODS AND SERVICE SALES

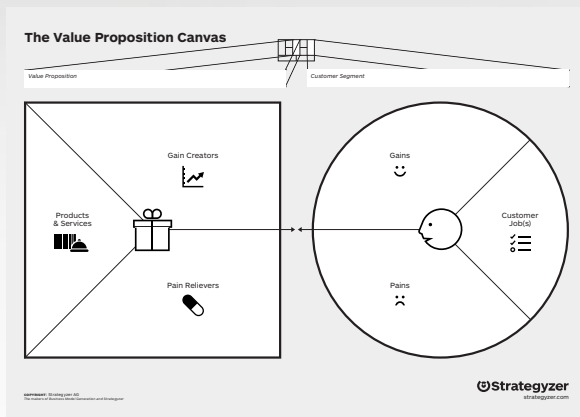
Key Aspect	Industrial Goods Sales	Industrial Service Sales	Key capabilities for selling industrial services
Underlying tenant	Persuasion model	Cocreation model	Ability to gain deep understanding of customer business model & KPI
Requirement Definition	Meet/exceed customer specifications	Discover/agree on result driven specifications	Ability to manage customer expectations
Network complexity	Limited number of stakeholders	Multiple stakeholders in customer & vendor organisations	Ability to develop strong networkties in customer and vendor organisations
Outcome orientation	Focus on closing opportunity (hunter perspective)	Focus on share growth & contract renewal (farmer perspective)	Ability to tangibilise outcomes from the customer perspective

SERVICE VALUE

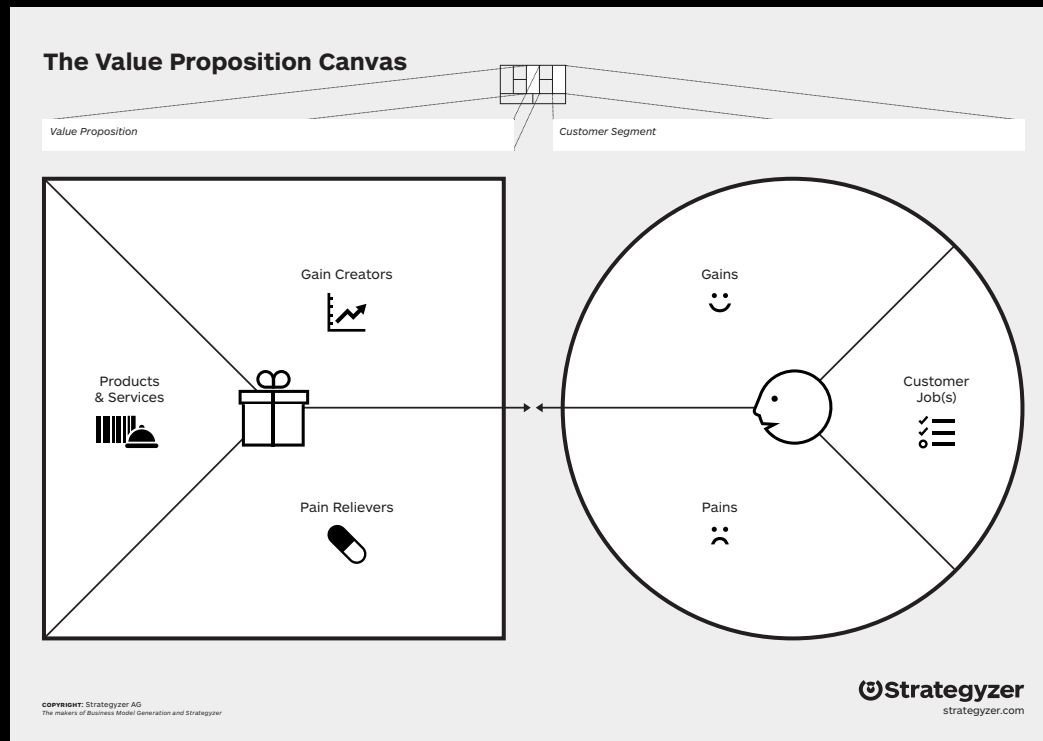


BUILDING A SERVICE CULTURE IN B2B: 4 STEPS





UNDERSTANDING THE VALUE



Describes the benefits customers can expect from your products and services.

Exercise 1 : What value does the product or service create for the user?

UNDERSTANDING THE BUSINESS MODEL

The Business Model Canvas

Designed for:

Designed by:

Date:

Version:

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
	Key Resources		Channels	
Cost Structure		Revenue Streams		

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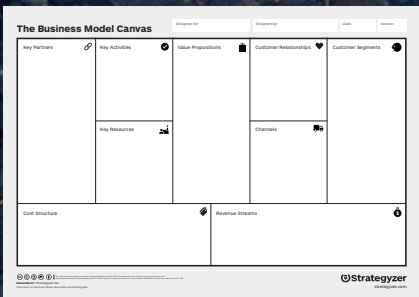
DESIGNED BY: Strategyzer AG
The makers of Business Model Generation and Strategyzer

Strategyzer
strategyzer.com

Rationale of how an organization creates, delivers and captures value.

Exercise 2

1. What elements do you need for the product or service?
2. Is there a business case here?



Over 1 Million

The number of elevators and escalators that we service around the world



200 TIMES PER SECOND

The number of times we measure the status of connected elevators



24/7

We monitor connected elevators around the clock. If a problem occurs, detailed information is available immediately



15,000

MEASUREMENTS A DAY

The amount of data a connected elevator sends to the cloud for analysis

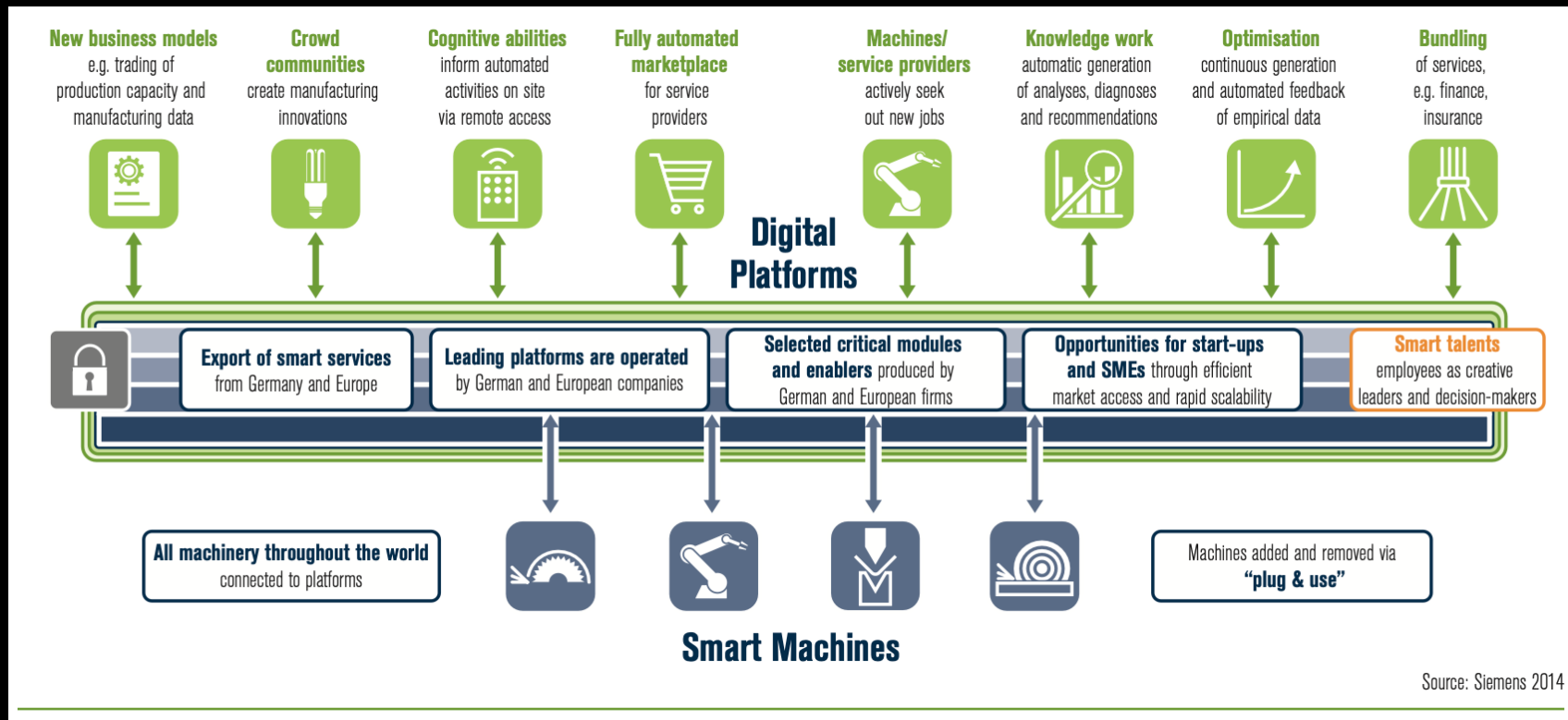


1 Billion

That's how many people we move every day with our products

5 FACTS ABOUT KONE'S 24/7 CONNECTED SERVICES

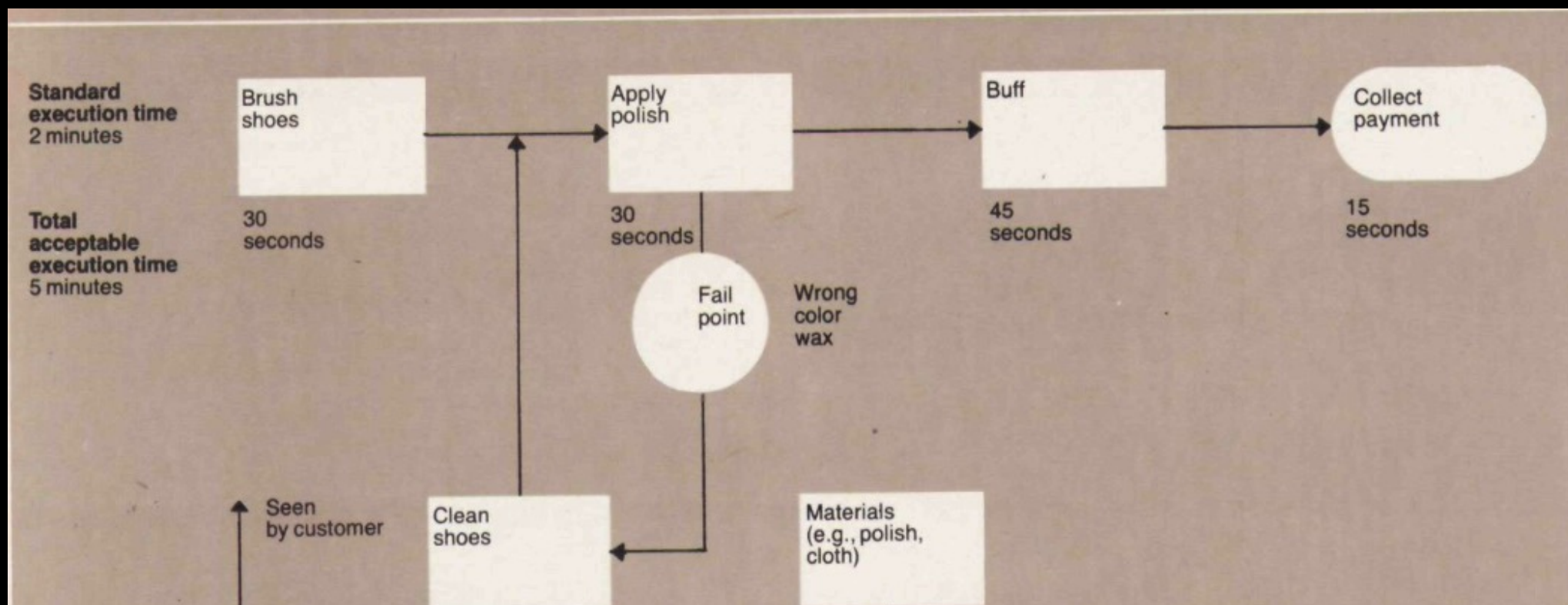
SMART MANUFACTURING SERVICES 2025



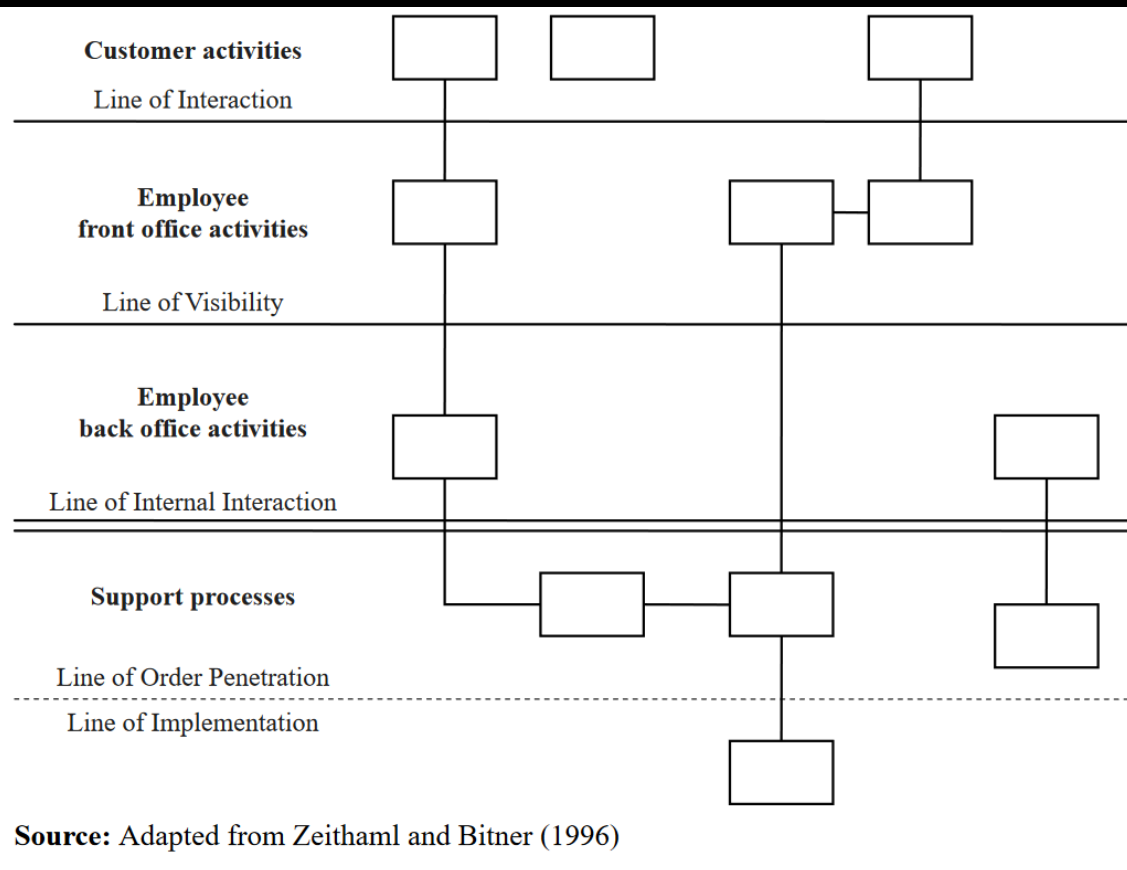
New Business Models can be created based on digital services

Based on smart products and digital platform.

SERVICE BLUEPRINT – SHOSTACK, G. LYNN, 1984



SERVICE BLUEPRINT



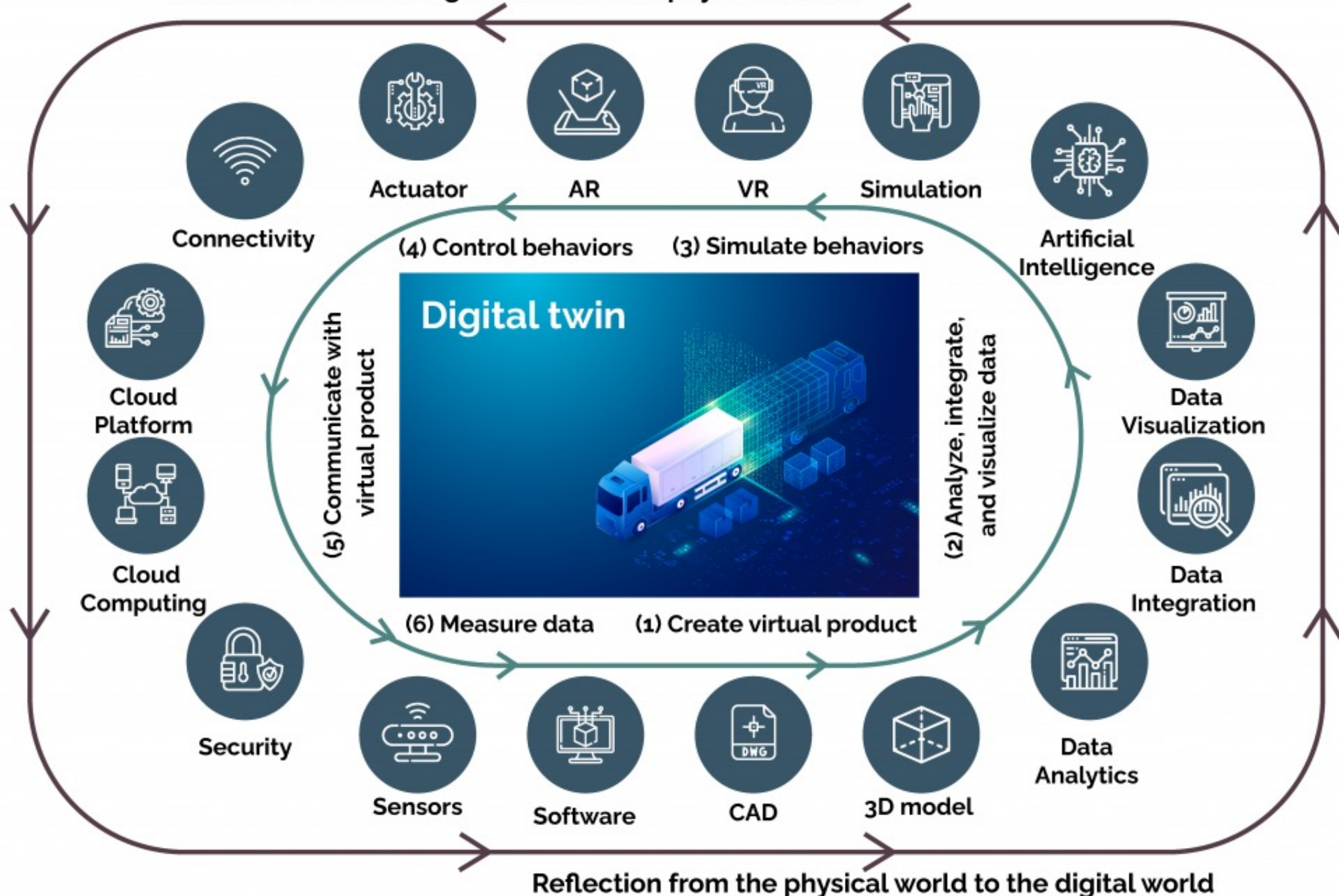
- Interactions along the lifecycle that require front office and backoffice activities.
- Importance increase when service are digital. CX become critical for service success and revenue.
- Examples: Rolls-Royce Power-by-the-Hour, Foodora, Amazon Web Services, eCommerce, Valmet Safe Harbour.

DEVELOPMENT PROCESS – SERVICE BLUEPRINT

	Aware	Join	Use	Develop	Leave
Product					
Customer Actions					
Onstage (visible employee actions)	The point where the user learns about the services	The sign-up or registration to the service	The usual usage of the service	The user's expanding usage of the service	The point when the user finishes using the service, either for a single session or forever
Backstage (invisible employee actions)					
Support Process					

- Service Blueprint pioneer G. Lynn Shostack for Citibank.
- Combination of understanding the customer journey and interaction of a service.
- Examples: banking service or any digital service.

Reflection from the digital world to the physical world





ANY
QUESTIONS
?

